3.2.2 Graphs

A8

Basic foundation content	Additional foundation content	Higher content only
work with coordinates in all four quadrants		

A9

Basic foundation content	Additional foundation content	Higher content only
plot graphs of equations that correspond to straight-line graphs in the coordinate plane	use the form $y = mx + c$ to identify parallel lines find the equation of the line through two given points, or through one point with a given gradient	use the form $y = mx + c$ to identify perpendicular lines

A10

Basic foundation content	Additional foundation content	Higher content only
identify and interpret gradients and intercepts of linear functions graphically and algebraically		

A11

Basic foundation content	Additional foundation content	Higher content only
	identify and interpret roots, intercepts and turning points of quadratic functions graphically	
	deduce roots algebraically	deduce turning points by completing the square

Notes: including the symmetrical property of a quadratic. See also A18

A12

Basic foundation content	Additional foundation content	Higher content only
recognise, sketch and interpret graphs of linear functions and quadratic functions	including simple cubic functions and the reciprocal function $y = \frac{1}{x}$ with $x \neq 0$	including exponential functions $y = k^x$ for positive values of k, and the trigonometric functions (with arguments in degrees) $y = \sin x$, $y = \cos x$ and $y = \tan x$ for angles of any size

Notes: see also G21

A13

Basic foundation content	Additional foundation content	Higher content only
		sketch translations and reflections of a given function

A14

Basic foundation content	Additional foundation content	Higher content only
plot and interpret graphs, and graphs of non-standard functions in real contexts, to find approximate solutions to problems such as simple kinematic problems involving distance, speed and acceleration	including reciprocal graphs	including exponential graphs

Notes: including problems requiring a graphical solution. See also A15

A15

Basic foundation content	Additional foundation content	Higher content only
		calculate or estimate gradients of graphs and areas under graphs (including quadratic and other non-linear graphs), and interpret results in cases such as distance-time graphs, velocity-time graphs and graphs in financial contexts

Notes: see also A14, R14 and R15

A16

Basic foundation content	Additional foundation content	Higher content only
		recognise and use the equation of a circle with centre at the origin find the equation of a tangent to a circle at a given point